MP1994V2 Proof Ledger

Mortensen–Pissarides (1994) Lean replication

31 July 2026

Purpose. The milestone tracker records project state. This cumulative ledger records exactly what Lean has proved, the hypotheses used, and the economic meaning of each theorem.

1 M1: Surplus Bellman equation

Paper reference	Section 3, journal page 401, equation (8)
Adequacy status	**COMPLETE – GREEN**
Lean source	`MP1994V2/SteadyState/Surplus.lean`

1.1 Mathematical statement

For the actual equilibrium surplus

$$S(\varepsilon) = J(\varepsilon) + W(\varepsilon) - U,$$

Equation (8) is a conditional representation theorem: for every object satisfying `ValueEquilibrium P`, Lean proves that the surplus derived from its value functions satisfies, for every idiosyncratic state ε,

$$(r+\lambda)S(\varepsilon) = p + \sigma\varepsilon - b + \lambda \int \max\{S(x), 0\}\, dF(x) - \beta\theta q(\theta)S(\varepsilon_u). \tag{8}$$

Here `P.shock` is the probability law of the idiosyncratic shock; its induced CDF is F, and integration with respect to dF is represented in Lean by integration against `P.shock`. Also, $\theta q(\theta)$ is `P.workerMeetingRate E.theta`. M1 does not prove existence of a `ValueEquilibrium`.

1.2 Economic interpretation

- $p + \sigma\varepsilon$: current match output.
- b: unemployment opportunity cost.
- $\lambda \int \max\{S(x), 0\}\, dF(x)$: option value of future idiosyncratic draws, allowing rejection of nonpositive continuation surplus.
- $\beta\theta q(\theta)S(\varepsilon_u)$: the employed worker's foregone search gain at the upper-support job.

1.3 Exact Lean declarations

Role	Declaration
Continuation decomposition	`MP1994V2.continuationTerm_eq_integral_activeSurplus_sub`
Firm surplus share	`MP1994V2.ValueEquilibrium.firm_share`
Algebraic surplus flow	`MP1994V2.ValueEquilibrium.surplus_flow_equation`
Minimal probability theorem	`MP1994V2.ValueEquilibrium.surplus_bellman_of_probability`
Paper-level wrapper	`MP1994V2.ValueEquilibrium.surplus_bellman`

1.4 Assumptions

Used:

- equation (3), through the definition of actual surplus;
- equation (4), only at the upper state;
- equations (5) and (6), added to cancel the wage;
- equation (7), subtracted to account for unemployment value;
- probability normalization of `P.shock`;
- `ValueEquilibrium.active_surplus_integrable`.

The minimal theorem assumes only `[IsProbabilityMeasure P.shock]`. The wrapper takes `D : ShockAssumptions P`, installs `D.isProbability` locally, and uses no other field of D.

Not used: parameter signs; `CoreEconomicAssumptions`; upper-support restrictions or maximality; atomlessness; mean zero; unit variance; `ShockNormalizationAssumptions`; matching positivity or monotonicity; `MatchingAssumptions`; equations (1) and (2).

1.5 Readable proof

Write the shared continuation term as

$$C(\varepsilon) = \int [\max\{S(x), 0\} - S(\varepsilon)]\ dF(x).$$

Adding equations (5) and (6) cancels the wage and sums the Nash weights:

$$r[J(\varepsilon) + W(\varepsilon)] = p + \sigma\varepsilon + \lambda C(\varepsilon).$$

Subtracting equation (7), and using equation (3), gives

$$rS(\varepsilon) = p + \sigma\varepsilon - b + \lambda C(\varepsilon) - \theta q(\theta)[W(\varepsilon_u) - U].$$

Equation (4) at ε_u gives

$$W(\varepsilon_u) - U = \beta S(\varepsilon_u).$$

Probability normalization of `P.shock` and active-surplus integrability give

$$C(\varepsilon) = \int \max\{S(x), 0\}\, dF(x) - S(\varepsilon).$$

Substitution and moving $-\lambda S(\varepsilon)$ to the left proves (8).

1.6 Adequacy and next dependency

The actual `ValueEquilibrium` surplus is used. Equation (8) is derived, not added as an assumption or field. No cutoff, reservation rule, or affine surplus result is used. It is conditional on an existing `ValueEquilibrium` and does not establish that such an equilibrium exists.

Human-review conclusion. Equation (8) matches the paper and is derived non-circularly from equations (3)–(7). The actual equilibrium surplus is used, with no cutoff, affinity result, or reduced equilibrium. The theorem is explicitly conditional on an existing `ValueEquilibrium`.

M2 uses (8) as the reviewed dependency for the affine-surplus and reservation results recorded next.

2 M2: Affine surplus and endogenous reservation cutoff

Paper reference	Section 3; journal p. 401, reservation-property paragraph; journal p. 402, equation (12)
Adequacy status	**COMPLETE – GREEN**
Lean sources	`MP1994V2/SteadyState/AffineSurplus.lean`; `MP1994V2/SteadyState/Cutoff.lean`

2.1 Mathematical statements

For every existing `ValueEquilibrium P`, subtracting equation (8) at two states gives

$$(r+\lambda)\,[S(\varepsilon_2) - S(\varepsilon_1)] = \sigma(\varepsilon_2 - \varepsilon_1).$$

Under $r > 0$, $\lambda \geq 0$, and $\sigma > 0$,

$$S(\varepsilon_2) - S(\varepsilon_1) = \frac{\sigma}{r+\lambda}(\varepsilon_2 - \varepsilon_1),$$

so S is strictly increasing. Define

$$\varepsilon_d = \varepsilon_u - \frac{(r+\lambda)S(\varepsilon_u)}{\sigma}.$$

Lean proves a unique zero on all of $\mathbb{R}$:

$$S(\varepsilon_d) = 0, \qquad S(\varepsilon) = 0 \iff \varepsilon = \varepsilon_d.$$

The reservation rules are

$$S(\varepsilon) < 0 \iff \varepsilon < \varepsilon_d, \quad 0 \leq S(\varepsilon) \iff \varepsilon_d \leq \varepsilon,$$
$$J(\varepsilon) < 0 \iff \varepsilon < \varepsilon_d, \quad 0 \leq J(\varepsilon) \iff \varepsilon_d \leq \varepsilon.$$

Paper equation (12) and its zero-anchored form are

$$S(\varepsilon) - S(\varepsilon_d) = \frac{\sigma(\varepsilon - \varepsilon_d)}{r+\lambda}, \qquad S(\varepsilon) = \frac{\sigma}{r+\lambda}(\varepsilon - \varepsilon_d).$$

The M3 bridge is

$$\max\{S(\varepsilon), 0\} = \frac{\sigma}{r+\lambda}\max\{\varepsilon - \varepsilon_d, 0\}.$$

2.2 Proof strategy and economic separation

The expectation and upper-job search term in equation (8) are common across current states, so they cancel when the two equation instances are subtracted. The resulting positive affine slope constructs the unique mathematical zero.

Economic admissibility is proved separately. Equation (2) and $r > 0$ give $V = 0$; equation (1) then gives

$$q(\theta)J(\varepsilon_u) = c.$$

Positive vacancy cost and meeting imply $J(\varepsilon_u) > 0$. Since $1 - \beta > 0$, the upper-state surplus is positive, and therefore $\varepsilon_d < \varepsilon_u$. Recruiting cost thus places the reservation state strictly below the productivity of a newly formed upper-support job.

2.3 Exact Lean declarations

Role	Declaration
Scaled two-point identity	`ValueEquilibrium.surplus_difference_scaled_of_probability`
Divided affine identity	`ValueEquilibrium.surplus_difference`
Strict monotonicity	`ValueEquilibrium.surplus_strictMono`
Derived cutoff	`ValueEquilibrium.reservationCutoff`
Cutoff zero	`ValueEquilibrium.surplus_reservationCutoff_eq_zero`
Unique zero	`ValueEquilibrium.existsUnique_surplus_zero`
Vacancy value	`ValueEquilibrium.vacancy_value_eq_zero`
Free-entry product	`ValueEquilibrium.vacancy_meeting_mul_upper_firm_value_eq_cost`
Upper firm value	`ValueEquilibrium.upper_firm_value_pos`
Upper surplus	`ValueEquilibrium.upper_surplus_pos`
Cutoff below upper point	`ValueEquilibrium.reservationCutoff_lt_epsUpper`
Equation (12)	`ValueEquilibrium.equation12`
Surplus sign rule	`ValueEquilibrium.surplus_neg_iff_lt_reservationCutoff`
Firm sign rule	`ValueEquilibrium.firm_value_neg_iff_lt_reservationCutoff`
Active-surplus bridge	`ValueEquilibrium.activeSurplus_eq_slope_mul_positivePart`
Capstone	`ValueEquilibrium.milestone2_capstone`

Capstone zero identification. The capstone explicitly includes

$$\forall\varepsilon, \quad S(\varepsilon) = 0 \iff \varepsilon = \varepsilon_d,$$

rather than merely asserting an unidentified unique zero.

2.4 Assumptions

Core bundle. The affinity and cutoff theorems formally take `A : CoreEconomicAssumptions P`. Within that bundle their proof terms access `r_pos`, `lambda_nonneg`, and `sigma_pos`. Economic admissibility additionally accesses `c_pos` and `beta_lt_one`, as well as equations (1)–(2) and equilibrium $\theta > 0$.

Matching bundle. Paper-facing admissibility and capstone theorems formally take

`M : MatchingAssumptions P.`

Their proof terms access only `vacancyMeetingRate_pos`. The monotonicity fields remain part of the bundled theorem premise even though they are not referenced by the proof term.

Shock bundle. Wrappers taking `D : ShockAssumptions P` access only `D.isProbability` in the M2 affinity and zero arguments. The remaining distributional fields are carried by the paper-level bundle but are not accessed by those proof terms.

Thus the formal paper-facing premises include the complete bundles where shown in the signatures. Within them, the proofs do not access atomlessness, upper-support maximality, shock moment normalization, matching monotonicity, or any elasticity or differentiability restriction.

2.5 Equation (12) dependency

`equation12` follows from the affine-difference theorem evaluated at the derived cutoff. The separate zero theorem is required to simplify (12) to

$$S(\varepsilon) = \frac{\sigma}{r+\lambda}(\varepsilon - \varepsilon_d).$$

2.6 Interpretation, limitations, and next dependency

The positive slope means that higher idiosyncratic productivity raises joint match surplus by the constant factor $\sigma/(r+\lambda)$. The unique zero is the reservation productivity: firms destroy matches below it, are indifferent at it, and retain matches at or above it.

These results are conditional on an existing `ValueEquilibrium`. M2 does not prove existence of that equilibrium or of the joint pair (θ, ε_d). It proves a unique zero on $\mathbb{R}$ and that the zero lies below ε_u, but not that it belongs to the topological support of the shock law. No lower-support assumption is present; the cutoff may lie below the lower effective support, making destruction probability zero.

Human-review conclusion. The surplus-affinity proof is non-circular, and the cutoff is derived from actual equilibrium values. Mathematical zero existence and economic admissibility remain separate; equations (1)–(2) are used only for admissibility. No M3 result is assumed or proved. The reviewed capstone explicitly identifies every surplus zero with `reservationCutoff`.

M3 may use the active-surplus positive-part formula to derive the continuation-value identity and the job-destruction and job-creation equations. No such integral identity is proved in M2.

3 M3: Continuation value, job destruction, and job creation

Paper reference	Section 3; journal page 401, equations (9)–(10); journal page 402, equation (13)
Adequacy status	**COMPLETE – GREEN**

Lean sources	`SteadyState/Continuation.lean`
	`SteadyState/JobDestruction.lean`
	`SteadyState/JobCreation.lean`

Capstone source	`SteadyState/StaticConditions.lean`

3.1 Definitions and continuation identity

The foundational expected excess and paper-facing tail option value are

$$H(d) = \int \max\{x-d, 0\}\, dP.\mathrm{shock}(x), \qquad T(d) = \int_d^{\varepsilon_u} [1-F(x)]\, dx.$$

The CDF F is induced by `P.shock`; it is not a separate primitive. Lean also defines measure and tail forms of the job-destruction residual, the paper-form job-creation residual, and zero predicates for those residuals.

M2 supplies

$$\max\{S(x), 0\} = \frac{\sigma}{r+\lambda} \max\{x - \varepsilon_d, 0\}.$$

Integrating gives the foundational measure-form continuation identity

$$\int \max\{S(x), 0\}\, dP.\mathrm{shock}(x) = \frac{\sigma}{r+\lambda} H(\varepsilon_d).$$

Integrability of the positive part is derived from `active_surplus_integrable` and the nonzero positive M2 slope; it is not added as a new assumption.

3.2 Strict-tail layer cake

Mathlib's `MeasureTheory.Integrable.integral_eq_integral_meas_lt` gives

$$H(\varepsilon_d) = \int_{t>0} P.\text{shock}\{x : t < \max(x - \varepsilon_d, 0)\}\, dt.$$

For $t > 0$, the level set is $(\varepsilon_d + t, \infty)$. Probability normalization proves

$$P.\text{shock.real}((y, \infty)) = 1 - F(y).$$

The almost-sure upper bound makes the tail vanish beyond ε_u. Truncation and the interval-translation lemma then give

$$H(\varepsilon_d) = T(\varepsilon_d).$$

The strict-tail theorem avoids the paper's informal differentiation and integration-by-parts wording. It also avoids atomlessness: no conversion between weak and strict tails is required.

3.3 Equations (9) and (10)

Substitution in equation (8) produces equation (9):

$$\begin{aligned}(r+\lambda)S(\varepsilon) = p + \sigma\varepsilon - b + \frac{\lambda\sigma}{r+\lambda}H(\varepsilon_d)\\ - \beta\theta q(\theta)S(\varepsilon_u),\end{aligned} \tag{9M}$$

and the paper form replaces $H(\varepsilon_d)$ by $T(\varepsilon_d)$.

Equations (1)–(2) and firm sharing give

$$\beta\theta q(\theta)S(\varepsilon_u) = \frac{\beta c}{1-\beta}\theta.$$

Evaluating (9M) at the zero-surplus cutoff gives the measure form

$$p + \sigma\varepsilon_d = b + \frac{\beta c}{1-\beta}\theta - \frac{\lambda\sigma}{r+\lambda}H(\varepsilon_d).$$

Replacing H by T is paper equation (10). The middle term is the worker's foregone search opportunity; the final subtraction is the option value of retaining a temporarily unprofitable job.

3.4 Equation (13)

Independently of equation (10), free entry, firm sharing, and M2 affine surplus at the new-job state give

$$q(\theta)(1-\beta)\frac{\sigma}{r+\lambda}(\varepsilon_u - \varepsilon_d) = c.$$

Positivity of the denominator factors yields

$$q(\theta) = \frac{c}{1-\beta}\frac{r+\lambda}{\sigma(\varepsilon_u - \varepsilon_d)}. \tag{13}$$

A smaller productivity gap raises the required vacancy meeting rate. Strict decrease of q implies lower market tightness, while the current weak monotonicity assumption on the worker meeting rate implies a weakly lower worker job-finding rate. M3 does not prove a comparative-static theorem across equilibria.

3.5 Exact Lean declarations

Role	Declaration
Expected excess	`Primitives.expectedExcess`
Tail probability	`Primitives.tailProbability`
Tail option value	`Primitives.tailOptionValue`
Strict-tail/CDF identity	`Primitives.measureReal_Ioi_eq_one_sub_cdf`
Excess integrability	`ValueEquilibrium.positivePart_sub_cutoff_integrable`
Layer-cake identity	`ValueEquilibrium.expectedExcess_cutoff_eq_tailOptionValue`
Active-surplus expectation	`ValueEquilibrium.integral_activeSurplus_eq_slope_mul_expectedExcess`
Equation (9), measure form	`ValueEquilibrium.equation9_measure`
Equation (9), paper form	`ValueEquilibrium.equation9`
Search gain	`ValueEquilibrium.search_gain_eq`
Equation (10), measure form	`ValueEquilibrium.equation10_measure`
Equation (10), paper form	`ValueEquilibrium.equation10`
Destruction residual, measure	`Primitives.jobDestructionResidualMeasure`
Destruction residual, tail	`Primitives.jobDestructionResidual`
Destruction witness, measure	`ValueEquilibrium.satisfiesJobDestructionMeasure`
Destruction witness, tail	`ValueEquilibrium.satisfiesJobDestruction`
Creation product	`ValueEquilibrium.job_creation_product_identity`
Equation (13)	`ValueEquilibrium.equation13`
Creation residual	`Primitives.jobCreationResidual`
Creation witness	`ValueEquilibrium.satisfiesJobCreation`
Capstone	`ValueEquilibrium.milestone3_capstone`

3.6 Assumptions and limitations

Paper-facing theorems formally carry the complete `CoreEconomicAssumptions`, `ShockAssumptions`, and `MatchingAssumptions` bundles.

- Measure-form continuation and equation (9) use probability normalization, active-surplus integrability, and the core fields `r_pos`, `lambda_nonneg`, and `sigma_pos`.
- The tail identity accesses `D.isProbability` and `D.upperSupport`. It does not access `D.noAtoms` or first-moment integrability.
- Its matching bundle is used through `reservationCutoff_lt_epsUpper`, whose proof accesses only `vacancyMeetingRate_pos`. Matching monotonicity remains in the formal bundle but is not referenced by the M3 proof term.
- Equation (10) additionally uses free entry and `beta_lt_one`. Its measure form requires no matching bundle.
- Equation (13) uses free entry, firm sharing, denominator positivity, and cutoff admissibility. Its shock bundle installs only probability normalization for the M2 affine formula.

Shock mean zero, unit variance, `ShockNormalizationAssumptions`, atomlessness, matching elasticity, differentiability, matching monotonicity, and comparative-static assumptions are deliberately unused.

M3 remains conditional on an existing `ValueEquilibrium`. It introduces no reduced-equilibrium structure, reverse reconstruction, joint-equilibrium existence or uniqueness, unemployment stock, or comparative-static theorem.

Human-review conclusion. The strict-tail layer-cake derivation is valid and does not use atomlessness. Equations (9), (10), and (13) match the paper, and equations (10) and (13) are derived independently. The residual results are forward conditional theorems for an existing `ValueEquilibrium`; no reduced equilibrium, reconstruction, or existence theorem is proved.

3.7 Residual admissibility warning

`SatisfiesJobDestruction` and `SatisfiesJobCreation` alone are not a complete reduced equilibrium. M4 must add $\theta > 0$, $d < \varepsilon_u$, and the relevant primitive, distributional, denominator, and other admissibility assumptions. The quotient-form job-creation residual must not be used without proving its denominator nonzero. The product-form job-creation identity is the robust foundational representation.

3.8 Dependency unlocked

M4 may define a `ReducedEquilibrium` only after supplying the missing admissibility contract. The current tail identity obtains integrability from an existing `ValueEquilibrium`. Reverse reconstruction requires a generic proof that $x \mapsto \max\{x - d, 0\}$ is integrable using the assumption

`ShockAssumptions.firstMomentIntegrable`.

M4 also needs, for an arbitrary admissible cutoff d, the generic identity

`expectedExcess` $d =$ `tailOptionValue` d,

not only the specialization at `E.reservationCutoff`, before constructing value functions from a reduced pair.

4 M4: Reduced-equilibrium reconstruction and equivalence

Paper reference	Section 3, equations (10), (13), journal page 402
Adequacy status	**COMPLETE – GREEN**
Lean sources	`Equilibrium/Reduced.lean`; `SteadyState/ReducedAnalytic.lean`; `ForwardBridge.lean`; `Reconstruction.lean`; `Equivalence.lean`

4.1 Objects and robust conditions

`ValueEquilibrium P` is the primitive value system. `ReducedEquilibrium P` stores only $\theta > 0$, $d < \varepsilon_u$, measure-form job destruction, and product-form job creation. A full steady state additionally contains unemployment and is deferred to M6. Measure form integrates directly against `P.shock`; product form avoids division by $\varepsilon_u - d$. The CDF-tail equation (10) and quotient equation (13) are derived.

4.2 Generic analysis and reconstruction

`Primitives.positivePart_sub_integrable` uses probability normalization and first-moment integrability. The generic expected-excess/tail theorem additionally uses the almost-sure upper bound to identify the induced-CDF tail integral for every $d \le \varepsilon_u$. Atomlessness is unused.

For a reduced pair R, Lean reconstructs

$$\begin{aligned} S_R(\varepsilon) &= \frac{\sigma}{r+\lambda}(\varepsilon - d), & V_R &= 0, & J_R &= (1-\beta)S_R, \\ U_R &= \frac{b + \beta\theta q(\theta) S_R(\varepsilon_u)}{r}, & W_R &= U_R + \beta S_R, \\ w_R(\varepsilon) &= \beta[p + \sigma\varepsilon] + (1-\beta)b + \beta c\theta. \end{aligned}$$

The wage is the Nash formula, not an arbitrary residual.

4.3 Verification and public declarations

Product-form job creation proves equation (1), while $V_R = 0$ proves (2). The computed surplus gives (3), worker values give (4), and the definition of U_R gives (7). Measure-form job destruction, expected excess, and search gain first derive candidate equation (8); continuation decomposition and the explicit wage then prove (5) and (6).

Principal declarations include:

- `Primitives.SatisfiesJobCreationProduct`; `ReducedEquilibrium`.
- `Primitives.positivePart_sub_integrable`; `Primitives.expectedExcess_eq_tailOptionValue`.
- `ValueEquilibrium.toReducedEquilibrium`; `ReducedEquilibrium.toValueEquilibrium`.
- `ReducedEquilibrium.toValue_toReduced`; `ValueEquilibrium.wage_eq_nash_formula`.
- `ValueEquilibrium.reconstruction_components`; `ValueEquilibrium.reconstruction_economic_roundtrip`.
- `valueEquilibrium_nonempty_iff_reducedEquilibrium_nonempty`; `m4_representation_capstone`.

4.4 Assumptions and exact scope

The forward bridge carries core, shock, and matching bundles; matching enters through M2 cutoff admissibility. Reverse reconstruction carries only core and shock bundles. The reverse constructor uses probability normalization and first-moment integrability, but not upper support. Upper support is needed for the paper-facing CDF-tail form of equation (10). Thus a full assumption bundle may occur in a signature while a proof term accesses only these named fields. Atomlessness, moment normalization, matching monotonicity, elasticity, and differentiability are unused.

The reduced round trip is exact. The value round trip proves equality of theta, V, U, J, W, wage, and surplus. Exact structure equality is not exported because proof-valued Bellman/admissibility fields are not the stable economic interface. Under the maintained bundles, nonemptiness is logically equivalent. Neither side is proved nonempty because no initial witness is constructed. M4 proves neither existence nor uniqueness.

Human-review conclusion. The robust reduced conditions and explicit Nash wage are economically correct. Equations (1)–(7) are verified rather than assumed; the two directions are non-circular; the reduced round trip is exact; and the value round trip preserves every economic object and surplus. The generic excess/tail analysis is valid, and no existence or uniqueness theorem is proved. M4 is **COMPLETE – GREEN**.

Dependency unlocked. M5 may study existence and uniqueness in the two-variable reduced system, knowing every admissible solution reconstructs the primitive value equilibrium.

5 M5: Existence and uniqueness of the static reduced equilibrium

Paper reference. Section 3, equations (10) and (13), Figure 1, and the uniqueness statement on journal page 402.

Adequacy status. Uniqueness: **COMPLETE – GREEN**. Existence: **COMPLETE – AMBER**. Overall M5: **COMPLETE – AMBER**.

5.1 Scalar curves and uniqueness

Define

$$\begin{aligned} H(d) &= \int \max\{x-d,0\}\,dP, \\ N_{JD}(d) &= p+\sigma d - b + \frac{\lambda\sigma}{r+\lambda}H(d), \\ \theta_{JD}(d) &= \frac{N_{JD}(d)}{\beta c/(1-\beta)}, \\ R(d) &= q(\theta_{JD}(d))\frac{(1-\beta)\sigma}{r+\lambda}(\varepsilon_u - d) - c. \end{aligned}$$

Positive part is one-Lipschitz, hence $|H(d_2)-H(d_1)| \le |d_2-d_1|$. Expected excess is therefore antitone and continuous. For $d_1 < d_2$, the increase in N_{JD} is bounded below by $\sigma r(d_2-d_1)/(r+\lambda) > 0$, so the JD-implied tightness is strictly increasing. Strict decrease and positivity of q make the positive-tightness JC locus strictly downward sloping. These opposite slopes prove `ReducedEquilibrium.unique` without an existence assumption.

5.2 Explicit existence correction

Slopes alone do not imply an intersection. M5 adds exactly:

> `StaticExistenceAssumptions.q_continuousOn_pos`: continuity of q on positive tightness; and `lower_crossing`: a $d_0 < \varepsilon_u$ with $\theta_{JD}(d_0) > 0$ and $R(d_0) > 0$.

This bundle contains no root or equilibrium witness. Lean derives $R(\varepsilon_u) = -c < 0$, proves bracket continuity, and applies Mathlib's `intermediate_value_Icc'` to obtain a strictly interior root. The root constructs a reduced equilibrium. Thus existence is proved under explicit additional existence conditions, not under the original primitive assumptions alone.

The lower crossing is not circular: it assumes a positive residual rather than a root. It nevertheless supplies the missing endpoint condition. Figure 1 proves uniqueness from slopes but does not itself prove existence, so M5 is a corrected conditional-existence theorem.

5.3 Existence assumption and primitive upgrade path

The exact committed closure bundle is:

```
structure StaticExistenceAssumptions
(P : Primitives) :  Prop where
q_continuousOn_pos :
ContinuousOn P.q (Set.Ioi 0)
lower_crossing :
∃ d0 : ℝ,
d0 < P.epsUpper ∧
0 < P.jobDestructionTheta d0 ∧
0 < P.staticCrossingResidual d0
```

`q_continuousOn_pos` is a regularity assumption: it makes the scalar residual continuous wherever JD-implied tightness is positive. `lower_crossing` is the substantive endpoint closure. It supplies the positive lower residual while Lean derives the upper residual as $-c < 0$. Because the lower sign is not currently derived from primitives, existence remains **COMPLETE – AMBER**.

A proposed, unformalized matching-boundary bundle would add $q(\theta) \longrightarrow +\infty$ as $\theta \downarrow 0$, together with continuity of q on positive tightness. This Inada condition is not sufficient by itself. The separate primitive profitability condition $b < p + \sigma\varepsilon_u$ is needed to ensure that the JD curve reaches positive tightness. If `jobDestructionNet` is nonpositive everywhere, behavior of q on positive tightness cannot create an equilibrium.

The proposed five-step route is:

1. use the one-Lipschitz bound on expected excess to obtain a sufficiently low cutoff with negative JD net value;
2. use upper support and upper-job profitability to show positive JD net value at ε_u;
3. use continuity and strict increase to obtain a unique zero-net, zero-tightness boundary cutoff below ε_u;
4. use the right-hand Inada limit to obtain a positive scalar residual just above that boundary; and
5. construct `StaticExistenceAssumptions` and reuse `reducedEquilibrium_nonempty A D X` unchanged.

For $q(\theta) = \kappa\theta^{-\eta}$, with $\kappa > 0$ and $0 < \eta < 1$, vacancy filling is positive, strictly decreasing, and Inada at zero, while $\theta q(\theta)$ increases. A future M5b can certify these facts with `Real.rpow`, combine them with upper-job profitability, and upgrade existence to green for that specification. No such theorem is currently implemented.

The full economic argument, comparison table, proposed bundles, and future Lean theorem targets are in `docs/static_existence_foundation.md`.

5.4 Principal declarations and transport

- `Primitives.expectedExcess_antitone`, `expectedExcess_lipschitz`, and `expectedExcess_continuous`.
- `Primitives.jobDestructionNet`, `jobDestructionTheta`, `staticCrossingResidual`, and `staticCrossingResidual_strictAntiOn`.
- `jobDestruction_curve_strictMono` and `jobCreation_curve_strictAnti`.
- `ReducedEquilibrium.unique` and `reducedEquilibrium_atMostOne`.
- `StaticExistenceAssumptions`, `exists_staticCrossing`, `reducedEquilibrium_nonempty`, and `reducedEquilibrium_existsUnique`.
- `valueEquilibrium_nonempty`, `ValueEquilibrium.theta_eq`, `ValueEquilibrium.reservationCutoff_eq`, and `ValueEquilibrium.economically_unique`.
- `m5_static_equilibrium_capstone`.

Uniqueness accesses core sign restrictions; shock probability normalization and first-moment integrability; and vacancy-meeting positivity and strict decrease. Existence-only theorems take the core, shock, and static-existence bundles, but no matching bundle. They access only shock probability and first-moment integrability plus `q_continuousOn_pos` and `lower_crossing`. Combined uniqueness capstones retain matching.

M5 does not access upper support, atomlessness, shock mean or variance normalization, worker-meeting monotonicity, elasticity, or differentiability.

M4 reconstruction transports reduced nonemptiness to `ValueEquilibrium` and proves economic equality of theta, cutoff, values, wage, and surplus. Exact equality of proof-valued equilibrium records is not claimed.

M5 does not prove unconditional existence, equation (14), unemployment or vacancy stocks, comparative statics, or dynamics.

Human-review conclusion. The uniqueness and IVT proofs passed review. Uniqueness is **COMPLETE – GREEN**; conditional existence and overall M5 are **COMPLETE – AMBER**.

Remaining foundational gap. Derive `lower_crossing` from more primitive boundary/range conditions on q, or certify it for a standard Cobb–Douglas matching function. Until then, existence remains conditional. This does not block M6.

Dependency unlocked. M6 may add unemployment flow balance and equation (14) to the uniquely determined static pair.

6 M6: Unemployment flow balance and the full static steady state

Paper reference	Section 3; journal p. 403, equation (14); journal p. 404, flow discussion; Figure 2
Adequacy status	**COMPLETE – AMBER**
Human-review grade	Stock completion/equation (14): GREEN; uniqueness: GREEN; transported existence and overall M6: AMBER
Lean sources	`SteadyState/Flows.lean`, `Equilibrium/SteadyState.lean`, `SteadyState/Unemployment.lean`, and `SteadyState/FullEquilibrium.lean`

6.1 Strict destruction rule and paper CDF

The economic rule is strict, $\varepsilon < d$. Lean therefore defines

$$L_<(d) = \text{shock}((-\infty, d)), \qquad s(d) = \lambda L_<(d), \qquad f(\theta) = \theta q(\theta).$$

The flows are $D(d, u) = s(d)(1 - u)$ and $C(\theta, u) = f(\theta)u$. Under `ShockAssumptions.noAtoms`, `Primitives.strictShockBelow_eq_cdf` proves $L_<(d) = F(d)$. Atomlessness is used at this bridge between the strict destruction event and the paper's CDF notation; it was not needed for M3's layer-cake identity. Employment and vacancies are derived, not stored:

$$n = 1 - u, \qquad v = \theta u.$$

6.2 Equation (14) and stock uniqueness

Flow balance is $s(d)(1-u) = f(\theta)u$. Positive tightness and vacancy meeting imply $f(\theta) > 0$; nonnegative λ implies $s(d) \geq 0$. Thus $s(d) + f(\theta) > 0$, and the unique real solution is

$$u^*(\theta, d) = \frac{s(d)}{s(d) + f(\theta)}.$$

Lean proves $0 \leq u^* < 1$, direct flow balance, and stock uniqueness through the biconditional theorem

`Primitives.flow_balance_iff_unemployment_eq`.

Replacing the strict mass by the CDF yields paper equation (14):

$$u = \frac{\lambda F(d)}{\lambda F(d) + \theta q(\theta)}. \tag{14}$$

The paper-facing theorem is `SteadyStateEquilibrium.equation14`.

6.3 Full state, exact representation, and flows

`SteadyStateEquilibrium P` extends the reduced pair only with unemployment, its bounds, and flow balance. It stores neither employment, vacancies, equation (14), nor a value equilibrium. Every reduced state is completed by `ReducedEquilibrium.toSteadyStateEquilibrium`. Exact public round trips are

```
ReducedEquilibrium.toSteady_toReduced
SteadyStateEquilibrium.toReduced_toSteady
steadyStateEquivReduced.
```

The full state recovers primitive values through `SteadyStateEquilibrium.toValueEquilibrium`, preserving theta and cutoff. The accounting theorems establish

$$C = \theta q(\theta)u = q(\theta)v, \qquad D = \lambda F(d)(1-u), \qquad C = D.$$

6.4 Zero separation and the literal ratio

The model permits $s(d) = 0$. Then equation (14) gives $u = 0$, and the derived identity gives $v = 0$; the literal ratio is undefined. Lean proves $v = \theta u$ definitionally for every state, but proves $v/u = \theta$ only under `HasPositiveSeparationAt d`, after deriving $0 < u < 1$ and $0 < v$. The declarations are `unemployment_pos_of_positiveSeparation`, `vacancies_div_unemployment_eq_theta`, and the zero-separation theorems.

6.5 Declarations, assumptions, and adequacy

- Strict mass/CDF: `Primitives.strictShockBelow` and `strictShockBelow_eq_cdf`.
- Hazards/flows: `jobSeparationRate`, `jobFindingRate`, `jobDestructionFlow`, `jobCreationFlow`, and `totalTransitionHazard_pos`.
- Stock formula: `steadyStateUnemployment`, `steadyStateUnemployment_flow_balance`, and `flow_balance_iff_unemployment_eq`.
- Full representation: `SteadyStateEquilibrium`, both exact round trips, and `steadyStateEquivReduced`.
- Uniqueness/existence: `SteadyStateEquilibrium.unique`, `steadyStateEquilibrium_nonempty`, `steadyStateEquilibrium_existsUnique`, and the separate `m6_stock_completion_capstone` and `m6_conditional_existence_capstone`.

Stock completion accesses only nonnegative lambda, atomlessness for the CDF bridge, and positive vacancy meeting at positive theta. It does not use discounting, dispersion, bargaining, vacancy cost, moment integrability, upper support, normalization, matching antitonicity/worker-rate monotonicity, elasticity, or differentiability. Full-state uniqueness inherits reduced uniqueness. Full-state nonemptiness is transported from M5 and accesses `StaticExistenceAssumptions` only through its reduced witness.

Stock completion, equation (14), flow equality, exact representation, and full-state uniqueness introduce no new closure premise and are GREEN. M6 introduces no new existence closure assumption. Its AMBER existence grade is inherited entirely through

```
StaticExistenceAssumptions.lower_crossing
⟶ reducedEquilibrium_nonempty
⟶ toSteadyStateEquilibrium
⟶ steadyStateEquilibrium_nonempty.
```

The separate conditional-existence capstone makes this inherited premise visible; overall M6 is COMPLETE – AMBER.

Interiority is separate. Zero separation implies $u = v = 0$, so $v = \theta u$ remains valid but the literal ratio is unavailable. Positive separation is required only for the ratio theorem, not for stock completion or existence.

Human review judged equation (14), accounting, zero separation, exact representation, and uniqueness faithful and non-circular. The detailed record is `docs/static_steady_state_adequacy.md`. A future primitive route would expose positive lambda, lower-tail richness/support, and cutoff admissibility. The M5b route combines right-hand Inada behavior of q with upper-job profitability; proving it upgrades both M5 and inherited M6 existence without changing the M6 architecture.

M6 proves no unconditional existence, Beveridge-curve slope or convexity, comparative static, equation (15), dynamic, or M7+ theorem.

Dependency unlocked. M7 may study static comparative statics; M9 may reuse the flow residual for equation (15). M5b remains the route for upgrading existence.